Random list of findings as raw material for arxiv paper

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- ($C(10,2)^2$ setup) *Fact groups that have a semantic structure are easier to learn than facts without semantic structure.* Experimental setting: dim=64, 45*45 subjects, each in relationship R0 with 45 objects that share its two f attributes and in relationship R1 with 45 objects that share its two g attributes. If the training set consists of (i) 20% of all "semantic" pairs (810 pairs each for R0/R1) and (ii) 2x810 random facts, then the semantic pairs are learnt well, the random facts only at 17.8%. If (i) 60% of all "semantic" pairs (2430 pairs each for R0/R1) and (ii) 2x2430 random facts, then the semantic pairs are learnt well, the random facts only at chance level. (one_epoch_v7_random_control.md) NEED TO ADD GENERALIZATION RESULTS

- ($C(10,2)^2$ setup) *The MLP does not directly apply affine transformations to its input vector. Instead, it "implements from scratch" two functions $A_k x + b_k$ (one for each relation k) that produce the correct behavior for the two relations.* Is this the only way for the MLP to implement the desired functionality? It cannot implement two different affine transformations directly, only one? Also related to the non-linearity of GELU although that is probably not a real problem. [Could show that a single relation gives rise to a straightforward affine transformation, but in that case attention may take over.] (affine_activation_fit.md)

- ($C(10,2)^2$ setup) *Neurons are partitioned by relation.* 85 R0-only neurons, 82 R1-only neurons, 7 shared neurons, 82 dead neurons. (neuron_activation_histograms.md)

- ($C(10,2)^2$ setup) *The attention output is, to a very good approximation, a constant vector per relation — a pure relation-identity signal superimposed on the subject embedding.* The 1.1-unit pre-activation shift between matching and non-matching neurons is produced primarily by head 1's contribution to the residual stream. fc1 projects the relation direction from head 1 onto each neuron's weight vector; the sign and magnitude of this projection determines whether the neuron is gated on (matching relation) or off (non-matching relation). The relation direction is almost perfectly orthogonal to the subject subspace. (attention_decomposition_v7.md, neuron_input_decomposition_v7.md)

- ($C(10,2)^2$ setup) *If the MLP is removed, the model implements the affine transformation through attention.* This doesn't work so well for dim=64, head_dim=16, but almost as well as for the MLP variant for dim=128, head_dim=32. (attention_only_1layer_comparison.md) NEED TO ADD GENERALIZATION RESULTS

- ($C(10,2)^2$ setup) *Mechanism used by attention-only.* Heads 0 and 1 swap their attention targets depending on the relation token. Under R0, Head 1 reads the subject and maps it

through OV_1 (specialized for f-type), while Head 0 reads the relation (contributing a constant bias). Under R1, the roles flip — Head 0 reads the subject and maps it through OV_0 (specialized for g-type). The relation token doesn't add information to the residual stream; it controls which linear map is applied to the subject by flipping the attention routing. The whole computation is nearly perfectly linear ($R^2 = 0.9996$), with softmax acting as a near-binary switch. (attention_only_1layer_mechanism.md)

- (ravfogel setup) *In the Ravfogel setup, the network chooses one of two completely different ways of implementing the affine transformation: relation gating (as with the previous structured subject/object setup) vs clustering subjects into two different groups.* This choice is made early on in training (epoch 1-3). For dim=64, only the clustered implementation emerges. For dim=128, relation gating is more frequent, but clustering also occurs.

- (ravfogel setup) *Mechanistic analysis of the subject clustering implementation of an affine transformation.* Subjects were clustered, per relation, into 2 clusters, based on MLP input or post-GELU activations, with almost identical results. The approximation by affine transformations per cluster (and relation) is much better than per relation ($R^2 = 0.87$ vs $R^2 = 0.64$). 98% of subjects that are in cluster A under R0 land in cluster B under R1 (and vice versa). The separating direction is nearly the same (cosine 0.88 between the two directions), but the clusters swap sides. The separating direction is PC1 of the MLP input space — cosine alignment 0.9997 with the first principal component. Within-cluster relation mechanism:

Cell	Active neurons
Group A, R0	96
Group A, R1	83
Group B, R0	91
Group B, R1	94

XOR gating. 87 neurons are shared by A/R0 and B/R1, 77 neurons are shared by A/R1 and B/R0, only 3 active neurons do not follow this pattern (contradicts table?). (ravfogel_cluster_analysis.md)

- (ravfogel setup) *The separation between clusters is created by the attention layer, then amplified by LayerNorm.* NEED TO DIGGER DEEPER ON THIS. (ravfogel_cluster_analysis.md)

- (ravfogel setup) *The choice between relation gating and subject clustering is a phase transition, not a gradual process, and the outcome depends on random initialization.* 10 seeds (42–51) were trained at dim=128 for 3 epochs. All reach 100% accuracy by epoch 2, but the underlying mechanism varies dramatically:

  - 4/10 seeds: strong relation gating (SNR > 3, $|d| > 2.5$, attention between-relation variance $> 87\%$, shared neurons ≤ 1)
  - 4/10 seeds: partial/forming gating (SNR 0.6–1.8)
  - 2/10 seeds: no gating (SNR < 0.2, shared neurons > 230)

  The two non-gating seeds (43, 45) were trained for 50 epochs and *never* develop gating — they converge to the same nonlinear demixing mechanism as dim=64 (final SNR $\sim$0.1, $R^2 \sim 0.62$). Their loss is flat from epoch 10 onward (2.1209), providing no gradient signal for a mechanism transition. At dim=64, all 10 seeds uniformly show no gating at any epoch (max SNR = 0.12 across all seeds and epochs).

- (ravfogel setup) *Replace GELU with ReLU.* Many small polytopes (convex regions defined by which neurons are above/below zero), each with a different exact linear map. Per-cluster

R^2 (as a measure of how well the affine transformation approximates functionality) ranges from 0.82 to 0.99, with larger clusters (90 entities) at the lower end and smaller clusters (50 entities) at the upper end. Every subject has a unique activation pattern across ReLU neurons. subject_emb $\rightarrow$ object_emb is essentially random ($R^2 \approx 0.04$). The raw token embeddings of subjects and their corresponding objects are not affinely related. PCA on MLP inputs: the top principal component explains only 2.4% of variance – no structure in the subject embeddings.

- (ravfogel setup) *Replace GELU with GeGLU.* dim=64 results are messy. For dim=128: relation-gated mechanism. However, within each relation, there is little entity-level structure. Every subject has a unique gate pattern.

- (usable for any setup) *Embedding probes to measure how much of the affine transformation is already learned in the embeddings.* Trains per-relation affine transformations on subject embedding to object embedding mapping (object embeddings are rows of the unembedding matrix). Works great for relation-gating, but not so clear how to use it for subject clustering.

2026-05-19

Most work on factual knowledge of language models has treated facts as isolated factoids. But most facts are integrated into the general web of knowledge that the LM has learned. For example, if programmer X is 70 years old, then it is more likely that he knows Cobol. If programmer Y is working on performance-critical systems, then she most likely knows C++. So (X,knows,Cobol) and (Y,knows,C++) are not isolated factoids, but supported by what we know about X and Y. This semantic component of factual knowledge – mostly neglected in prior work – is what we study in this paper using a synthetic data methodology.

- It has been observed that some relations are represented linearly in language models, but there has been no clear explanation so far for this linearity. We show that semantically embedded relations give rise to linear relation representations whereas isolated facts are not stored linearly.

- We show that whether a linear representation emerges over the course of training depends on at least three factors: the density of the semantic connections between knowledge components, the amount of training data information on the semantic connections and the level of noise. The dependence on training data information elucidates one aspect of scaling laws for training data: a generalized representation of knowledge (such as a linear representation of a relation) only emerges at scale.

- The precise mechanism used by language models for realizing the linear representation of relations has been elusive. We show that MLPs can represent linear relations. The MLP's neurons' output vectors act as a basis. This basis can be frozen after initialization, with a minor effect on the quality of the final representation learned. The affine transformation corresponding to a relation is learned as a linear combination of the output vectors of the MLP, where only a subset is used for any particular relation. Each subject produces a different set of weights, i.e., the activation values of the neurons (this also depends on the relation, of course). Interestingly, this makes heavy use of the negative range of GELU: most of the weights for the relation-specific output vector subset are negative. The rank of the matrix of activations for the subjects closely corresponds to the number of subjects.

- An important practical question is under what circumstances a fact can be learned in pretraining. The most studied factor is frequency. We show that semantically embedded facts are reliably learned in pretraining after a single exposure (i.e., frequency 1). Thus, semantic embedding is at least as important as frequency for factual acquisition.

- The literature on factual knowledge has not addressed the problem of generalization. At first, this seems a natural choice: a fact is a fact because we know it to be one not because we are generalizing from other knowledge. However, consider the following biography: "After joining IBM in 1979, John spent most of his career working on banking systems for mainframe clients. He became known for understanding decades-old enterprise code that handled millions of records reliably every day." Arguably, given extensive knowledge about programmers and programming language, a model should predict (John,knows,Cobol). For the first time, we investigate how well models generalize to facts that are neither known nor logically deducible and show that generalization depends on linearity: if a linear representation has been learned for a relation, then it generalizes to new facts, otherwise there is no generalization.

- How does the successful acquisition of linear relation representations depend on model capacity, i.e., the number of parameters? We show that a minimum capacity is necessary (small models fail to learn the relation), but that with too much capacity, the model simply memorizes individual facts and does not learn to generalize. We also show that the standard training regime may not find the linear representation in cases where the capacity barely is sufficient; however, in such cases an auxiliary linearity objective can nudge learning towards successful learning of linearity and successful generalization.